

CHAPTER 16

NONLINEAR RESPONSE HISTORY ANALYSIS

16.1 GENERAL REQUIREMENTS

16.1.1 Scope. It shall be permitted to use nonlinear response history analysis in accordance with the requirements of this chapter to demonstrate acceptable strength, stiffness, and ductility to resist maximum considered earthquake (MCE_R) shaking with acceptable performance. When nonlinear response history analysis is performed, the design shall also satisfy the requirements of Section 16.1.2. Nonlinear response history analysis shall include the effects of horizontal motion, and where required by Section 16.1.3, vertical motion. Documentation of the design and analysis shall be prepared in accordance with Section 16.1.4. Ground motion acceleration histories shall be selected and modified in accordance with the procedures of Section 16.2. The structure shall be modeled and analyzed in accordance with the criteria in Section 16.3. Analysis results shall meet the acceptance criteria of Section 16.4. Independent structural design review shall be performed in accordance with the requirements of Section 16.5.

16.1.2 Linear Analysis. In addition to nonlinear response history analysis, a linear analysis in accordance with one of the applicable procedures of Chapter 12 shall also be performed. The structure's design shall meet all applicable criteria of Chapter 12. Where soil-structure interaction in accordance with Chapter 19 is used in the nonlinear analysis, it shall be permitted to also use the corresponding spectral adjustment in the linear analysis.

EXCEPTIONS:

1. For Risk Category I, II, and III structures, Sections 12.12.1 and 12.12.5 do not apply to the linear analysis. Where mean computed drifts from the nonlinear analyses exceed 150% of the permissible story drifts per Section 12.12.1, deformation-sensitive nonstructural components shall be designed for 2/3 of these mean drifts.
2. The overstrength factor, Ω_0 , is permitted to be taken as 1.0 for the seismic load effects of Section 12.4.3.
3. The redundancy factor, ρ , is permitted to be taken as 1.0.
4. Where accidental torsion is explicitly modeled in the nonlinear analysis, it shall be permitted to take the value of A_x as unity in the Chapter 12 analysis.

16.1.3 Vertical Response Analysis. Nonlinear response history analysis shall explicitly include the effects of vertical response where any of the following occur:

1. Vertical elements of the gravity force-resisting system are discontinuous.
2. For nonbuilding structures, when Chapter 15 requires consideration of vertical earthquake effects.

16.1.4 Documentation. Before performing the nonlinear analysis, project-specific design criteria shall be approved by the independent structural design reviewer(s) and the Authority Having Jurisdiction. The project-specific criteria shall identify the following:

1. The selected seismic and gravity force-resisting systems and procedures used in the structural design.
2. Geotechnical parameters including soil characteristics, recommended foundation types, design parameters, seismic hazard evaluation, target spectra, and selection and scaling of acceleration histories.
3. Design loading, including gravity and environmental loads.
4. Analytical modeling approach and assumptions, including software to be used, definition of mass, identification of force-controlled and deformation-controlled behaviors, description of which component actions are modeled elastically and inelastically, expected material properties, basis for hysteretic component modeling, component initial stiffness assumptions, joint stiffness assumptions, diaphragm modeling, damping, and procedure for modeling foundation-soil interaction.
5. Summaries of laboratory test data and other applicable data used to justify the hysteretic component modeling or used to justify acceptable structural performance.
6. Specific acceptance criteria values used for evaluating performance of elements of the seismic force-resisting system. Associated documentation shall also include identification of component failure modes deemed indicative of collapse.
7. Where drifts exceed 150% of the applicable values permitted in Section 12.12, the criteria used to demonstrate acceptable deformation compatibility of components of the gravity force-resisting system.

Following completion of analysis, the following documentation shall be prepared and presented to the independent structural design reviewer(s) and the Authority Having Jurisdiction:

1. Final geotechnical report, including soil shear strength, stiffness, and damping characteristics; recommended foundation types and design parameters; and seismic hazard evaluation, including both the target spectra and selection and scaling of ground motions.
2. Overall building dynamic behavior, including natural frequencies, mode shapes, and modal mass participation.
3. Key structural response parameter results and comparisons with the acceptance criteria of Section 16.4.
4. Detailing of critical elements.

16.2 GROUND MOTIONS

16.2.1 Target Response Spectrum. A target, 5%-damped, MCE_R response spectrum shall be developed using either the

procedures of Section 16.2.1.1 or Section 16.2.1.2. It shall be permitted to consider the effects of base slab averaging and foundation embedment in accordance with Chapter 19.

Where the effects of vertical earthquake shaking are included in the analysis, a target MCE_R vertical spectrum shall also be constructed.

16.2.1.1 Method 1. A single response spectrum shall be developed, based on the requirements of either Section 11.4.7 or Section 11.4.8.

16.2.1.2 Method 2. Two or more site-specific target response spectra shall be developed. When this method is used, the following requirements shall be fulfilled, in addition to the other requirements of this chapter:

1. Two or more periods shall be selected, corresponding to those periods of vibration that significantly contribute to the inelastic dynamic response of the building in two orthogonal directions. In the selection of periods, lengthening of the elastic periods of the model shall be considered.
2. For each selected period, a target spectrum shall be created that either matches or exceeds the MCE_R value at that period. When developing the target spectrum (1) site-specific disaggregation shall be performed to identify earthquake events that contribute most to the MCE_R ground motion at the selected period and (2) the target spectrum shall be developed to capture one or more spectral shapes for dominant magnitude and distance combinations revealed by the disaggregation.
3. The envelope of the target spectra shall not be less than 75% of the spectral values computed using Method 1 of Section 16.2.1.1, for all periods in the range specified in Section 16.2.3.1.
4. For each target response spectrum, a ground motion suite for response history analyses shall be developed and used in accordance with Sections 16.2.3 through Section 16.2.4. The acceptance criteria requirements of Section 16.4 shall be evaluated independently for each of the ground motion suites.

Variations on the procedures described in this section are permitted to be used when approved by the design review.

16.2.2 Ground Motion Selection. A suite of not less than 11 ground motions shall be selected for each target spectrum. Ground motions shall consist of pairs of orthogonal horizontal ground motion components and, where vertical earthquake effects are considered, a vertical ground motion component. Ground motions shall be selected from events within the same general tectonic regime and having generally consistent magnitudes and fault distances as those controlling the target spectrum and shall have similar spectral shape to the target spectrum. For near-fault sites, as defined in Section 11.4.1, and other sites where MCE_R shaking can exhibit directionality and impulsive characteristics, the proportion of ground motions with near-fault and rupture directivity effects shall represent the probability that MCE_R shaking will exhibit these effects. Where the required number of recorded ground motions is not available, it shall be permitted to supplement the available records with simulated ground motions. Ground motion simulations shall be consistent with the magnitudes, source characteristics, fault distances, and site conditions controlling the target spectrum.

16.2.3 Ground Motion Modification. Ground motions shall either be amplitude-scaled in accordance with the requirements of Section 16.2.3.2 or spectrally matched in accordance with the

requirements of section 16.2.3.3. Spectral matching shall not be used for near-fault sites unless the pulse characteristics of the ground motions are retained after the matching process has been completed.

16.2.3.1 Period Range for Scaling or Matching. A period range shall be determined, corresponding to the vibration periods that significantly contribute to the building's lateral dynamic response. This period range shall have an upper bound greater than or equal to twice the largest first-mode period in the principal horizontal directions of response, unless a lower value, not less than 1.5 times the largest first-mode period, is justified by dynamic analysis under MCE_R ground motions. The lower bound period shall be established such that the period range includes at least the number of elastic modes necessary to achieve 90% mass participation in each principal horizontal direction. The lower bound period shall not exceed 20% of the smallest first-mode period for the two principal horizontal directions of response. Where vertical response is considered in the analysis, the lower bound period used for modification of vertical components of ground motion need not be taken as less than the larger of 0.1 seconds, or the lowest period at which significant vertical mass participation occurs.

16.2.3.2 Amplitude Scaling. For each horizontal ground motion pair, a maximum-direction spectrum shall be constructed from the two horizontal ground motion components. Each ground motion shall be scaled, with an identical scale factor applied to both horizontal components, such that the average of the maximum-direction spectra from all ground motions generally matches or exceeds the target response spectrum over the period range defined in Section 16.2.3.1. The average of the maximum-direction spectra from all the ground motions shall not fall below 90% of the target response spectrum for any period within the same period range. Where vertical response is considered in the analysis, the vertical component of each ground motion shall be scaled such that the average of the vertical response spectra envelopes the target vertical response spectrum over the period range specified in Section 16.2.3.1.

16.2.3.3 Spectral Matching. Each pair of ground motions shall be modified such that the average of the maximum-direction spectra for the suite equals or exceeds 110% of the target spectrum over the period range defined in Section 16.2.3.1. Where vertical response is considered in the analysis, the vertical component of each ground motion shall be spectrally matched to the target vertical response spectrum such that the average of the matched spectra does not fall below the target vertical spectrum in the scaling range of Section 16.2.3.1.

16.2.4 Application of Ground Motions to the Structural Model. Ground motions shall be applied to the supports of the structural model. For near-fault sites, as defined in Section 11.4.1, each pair of horizontal ground motion components shall be rotated to the fault-normal and fault-parallel directions of the causative fault and applied to the building in such orientation. At all other sites, each pair of horizontal ground motion components shall be applied to the building at orthogonal orientations such that the average (or mean) of the component response spectrum for the records applied in each direction is within $\pm 10\%$ of the mean of the component response spectra of all records applied for the period range specified in Section 16.2.3.1.

16.3 MODELING AND ANALYSIS

16.3.1 Modeling. Mathematical models shall be three-dimensional and shall conform to the requirements of Section

12.7 and this section. For structures that have subterranean levels, the structural model shall extend to the foundation level and ground motions shall be input at the foundation level. All elements that significantly affect seismic response when subjected to MCE_R ground motions shall be included. Modeling of element nonlinear hysteretic behavior shall be consistent with ASCE 41 or applicable laboratory test data. Test data shall not be extrapolated beyond tested deformation levels. Degradation in element strength or stiffness shall be included in the hysteretic models unless it can be demonstrated that response is not sufficient to produce these effects.

Analysis models shall be capable of representing the flexibility of floor diaphragms where this is significant to the structure's response. Diaphragms at horizontal and vertical discontinuities in lateral resistance shall be explicitly modeled in a manner that permits capture of the force transfers and resulting deformations.

16.3.2 Gravity Load. The modeling of and demands on elements in the analysis model shall be determined considering earthquake effects acting in combination with expected gravity loads, both with and without live load. Expected gravity loads with live load shall be taken as $1.0D + 0.5L$, where L shall be taken as 80% of unreduced live loads that exceed 100 lb/ft^2 (4.79 kN/m^2) and 40% of all other unreduced live loads. Expected gravity loads without live load shall be taken as $1.0D$.

EXCEPTION: Where the sum, over the entire structure, of the expected live load ($0.5L$) as defined above does not exceed 25% of the total dead load, D , and the live load intensity L_0 over at least 75% of the structure is less than 100 psf (4.79 kN/m^2), the case without live load need not be considered.

16.3.3 P-Delta Effects. P-delta effects considering the spatial distribution of gravity loads shall be included in the analysis.

16.3.4 Torsion. Inherent eccentricity resulting from any offset in the centers of mass and stiffness at each level shall be accounted for in the analysis. In addition, where a Type 1a or Type 1b horizontal structural irregularity exists as defined in Section 12.3.2.1, accidental eccentricity consisting of an assumed displacement of the center of mass each way from its actual location by a distance equal to 5% of the diaphragm dimension of the structure parallel to the direction of mass shift shall be considered. The required 5% displacement of the center of mass need not be applied in both orthogonal directions at the same time.

16.3.5 Damping. Hysteretic energy dissipation of structural members shall be modeled directly. Additional inherent damping not associated with inelastic behavior of elements shall be modeled appropriate to the structure type and shall not exceed 2.5% equivalent viscous damping in the significant modes of response.

16.3.6 Explicit Foundation Modeling. When soil spring and/or dashpot elements are included in the structural model, horizontal input ground motions shall be applied to the horizontal soil elements rather than being applied to the foundation directly.

16.4 ANALYSIS RESULTS AND ACCEPTANCE CRITERIA

Structures shall be demonstrated to meet the global acceptance criteria of Section 16.4.1 and the element-level acceptance criteria of Section 16.4.2.

The mean value of story drift, and element demand, Q_u , shall be used to evaluate acceptability.

EXCEPTION: Where a ground motion produces unacceptable response as permitted in Section 16.4.1.1, 120% of the median value, but not less than the mean value obtained from the suite of analyses producing acceptable response shall be used.

16.4.1 Global Acceptance Criteria

16.4.1.1 Unacceptable Response. Unacceptable response to ground motion shall consist of any of the following:

1. Analytical solution fails to converge,
2. Predicted demands on deformation-controlled elements exceed the valid range of modeling,
3. Predicted demands on critical or ordinary force-controlled elements, as defined in Section 16.4.2 exceed the element capacity, or
4. Predicted deformation demands on elements not explicitly modeled exceed the deformation limits at which the members are no longer able to carry their gravity loads.

Unacceptable response to ground motion shall not be permitted.

EXCEPTION: For Risk Category I and II structures, where spectral matching of ground motion is not used, not more than one motion shall be permitted to produce unacceptable response.

16.4.1.2 Story Drift. The mean story drift ratio shall not exceed two times the limits of Table 12.12-1. The story drift ratio shall be computed as the largest difference of the deflections of vertically aligned points at the top and bottom of the story under consideration along any of the edges of the structure. For masonry shear wall structures, the limits of Table 12.12-1 applicable to masonry structures shall not apply and these structures shall instead comply with the limits for other structures.

16.4.2 Element-Level Acceptance Criteria. All element actions shall be evaluated either as force-controlled or deformation-controlled.

For each element action, the quantity Q_u shall be computed. Q_u shall be taken as the mean value of the response parameter of interest obtained from the suite of analyses.

Force-controlled response parameters shall be evaluated for acceptability in accordance with Section 16.4.2.1. Deformation-controlled response parameters shall be evaluated for acceptability in accordance with Section 16.4.2.2. Where required by Section 16.4.2.1 or Section 16.4.2.2, element actions shall be categorized as Critical, Ordinary, or Noncritical.

16.4.2.1 Force-Controlled Actions. Force-controlled actions shall satisfy Eq. (16.4-1):

$$\gamma I_e (Q_u - Q_{ns}) + Q_{ns} \leq Q_e \quad (16.4-1)$$

where I_e is the Importance Factor prescribed in Section 1.5.1; Q_{ns} is the demand caused by loads other than seismic; Q_e is the expected component strength; and γ is the load factor obtained from Table 16.4-1. Where an industry standard referenced in

Table 16.4-1 Load Factor for Force-Controlled Behaviors

Action Type	γ
Critical	2.0
Ordinary	1.5
Noncritical	1.0

Table 16.4-2 Seismic Resistance Factors for Critical and Ordinary Deformation-Controlled Actions

Action Type	ϕ_s
Critical	$0.3/I_e$
Ordinary	$0.5/I_e$

Chapter 14 defines expected strength, that value shall be used. Where this is not defined, it shall be permitted to calculate expected strength as the nominal strength defined in industry standards, except that expected material properties as defined in ASCE 41 shall be used in lieu of specified values.

EXCEPTIONS:

1. Noncritical force-controlled actions that are modeled, including consideration of strength loss effects, need not satisfy Eq. (16.4-1).
2. For actions other than shear in structural walls and columns, the nominal element strength need not exceed the effects of gravity load plus the force demand determined by plastic mechanism analysis, where the analysis is based on expected material properties.

16.4.2.2 Deformation-Controlled Actions. Deformation-controlled element actions shall not exceed the permissible acceptance criteria for the Collapse Prevention Structural Performance Level, as defined in ASCE 41, divided by the Importance Factor, I_e , prescribed in Section 1.5.1.

Alternatively, for critical and ordinary deformation-controlled actions, the mean inelastic deformation, Q_u , shall not exceed $\phi_s Q_{ne}$, where ϕ_s is obtained from Table 16.4-2 and Q_{ne} is the mean value of inelastic deformation at which loss of gravity-load-carrying capacity is expected to occur, based on laboratory test data. For noncritical deformation-controlled actions, inelastic deformation shall not exceed the valid range of modeling.

16.4.2.3 Elements of the Gravity Force-Resisting System. Elements that are not part of the seismic force-resisting system shall be demonstrated to be capable of supporting gravity loads using the mean building displacements from the suite of nonlinear response history analyses.

16.5 DESIGN REVIEW

Independent structural design review shall be performed in accordance with the requirements of this section. Upon completion of the review, the reviewer(s) shall provide the Authority Having Jurisdiction and the registered design professional with a letter attesting to

1. Scope of review performed,
2. Whether the reviewer(s) concur with the analysis and its applicability to the design,
3. Conformance of the design to applicable requirements of the standard, and
4. Any items relating to the design or analysis that require further resolution by the Authority Having Jurisdiction.

16.5.1 Reviewer Qualifications. Reviewer(s) shall consist of one or more individuals acceptable to the Authority Having Jurisdiction and possessing knowledge of the following items:

1. The requirements of this standard and the standards referenced herein, as they pertain to design of the type of structure under consideration.
2. Selection and scaling of ground motions for use in nonlinear response history analysis.
3. Analytical structural modeling for use in nonlinear response history analysis, including use of laboratory tests in the creation and calibration of the structural analysis models, and including knowledge of soil-structure interaction if used in the analysis or the treatment of ground motions.
4. Behavior of structural systems, of the type under consideration, when subjected to earthquake loading.

At least one reviewer shall be a registered design professional.

16.5.2 Review Scope. The scope of review shall include the items identified in Section 16.1.4, as well as the associated project documentation that demonstrates conformance to the design criteria.

16.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.